

Exercise JA

Q1. The integral $\int \frac{\sec^2 x}{(\sec x + \tan x)^{9/2}} dx$ equals (for some arbitrary constant K)

(A) $-\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7}(\sec x + \tan x)^2 \right\} + K$

(B) $\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7}(\sec x + \tan x)^2 \right\} + K$

(C) $-\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7}(\sec x + \tan x)^2 \right\} + K$

(D) $\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7}(\sec x + \tan x)^2 \right\} + K$

Ans. C

Sol. -

Q2. The value of the integral $\int_{-\pi/2}^{\pi/2} \left(x^2 + \ln \frac{\pi+x}{\pi-x} \right) \cos x$ is :

(A) 0 (B) $\frac{\pi^2}{2} - 4$ (C) $\frac{\pi^2}{2} + 4$ (D) $\frac{\pi^2}{2}$

Ans. B

Sol. -

Q3. For $a \in \mathbb{R}$ (the set of all real numbers), $a \neq -1$,

$$\lim_{n \rightarrow \infty} \frac{(1^\alpha + 2^\alpha + \dots + n^\alpha)}{(n+1)^{\alpha-1}[(n\alpha+1) + (na+2) + \dots + (na+n)]} = \frac{1}{60} \text{ Then } a =$$

(A) 5 (B) 7 (C) $\frac{-15}{2}$ (D) $\frac{-17}{2}$

Ans. B

Sol. -

Q4. Let $f : [a, b] \rightarrow [1, \infty)$ be a continuous function and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$g(x) = \begin{cases} 0 & \text{if } x < a, \\ \int_a^x f(t) dt & \text{if } a \leq x \leq b, \\ \int_a^b f(t) dt & \text{if } x > b \end{cases} \text{ Then}$$

(A) $g(x)$ is continuous but not differentiable at a (B) $g(x)$ is differentiable on $\mathbb{R}$

(C) $g(x)$ is continuous but not differentiable at b

(D) $g(x)$ is continuous and differentiable at either a or b but not both

Ans. A, C

Sol. -

Q5. The value of $\int_0^1 4x^3 \left\{ \frac{d^2}{dx^2} (1 - x^2)^5 \right\} dx$ is

Ans. 2

Sol. -

Q6. The following integral $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (2 \operatorname{cosec} x)^{17} dx$ is equal to :

- (A) $\int_0^{\log(1+\sqrt{2})} 2(e^u + e^{-u})^{16} du$ (B) $\int_0^{\log(1+\sqrt{2})} (e^u + e^{-u})^{17} du$ (C) $\int_0^{\log(1+\sqrt{2})} (e^u - e^{-u})^{17} du$
(D) $\int_0^{\log(1+\sqrt{2})} 2(e^u - e^{-u})^{16} du$

Ans. A

Sol. -

Q7. Let $f: [0,2] \rightarrow \mathbb{R}$ be a function which is continuous on $[0,2]$ and is differentiable on $(0,2)$ with $f(0) = 1$. Let $F(x) = \int_0^{x^2} f(\sqrt{t}) dt$ for $x \in [0,2]$. If $F'(x) = f(x)$ for all $x \in (0, 2)$, then $F(2)$ equals

- (A) $e^2 - 1$ (B) $e^4 - 1$ (C) $e - 1$ (D) e^4

Ans. B

Sol. -

Paragraph

Let there are n bags numbered 1 to n containing some white balls and remaining black balls. The i^{th} bag contains ' i ' balls. If probability of drawing a white ball from i^{th} bag is equal to $k i^2$ then answer the following questions.

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Q8. The value of $g\left(\frac{1}{2}\right)$ is

- (A) π (B) 2π (C) $\pi/2$ (D) $\pi/4$

Ans. A

Sol. -

Paragraph

Given that for each $a \in (0,1)$, $\lim_{h \rightarrow 0^+} \int_h^{1-h} (1-t)^{a-1} dt$ exists. Let this limit be $g(a)$. In addition, it is given that the function $g(a)$ is differentiable on $(0, 1)$.

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Q9. The value of $g'\left(\frac{1}{2}\right)$ is

- (A) $\pi/2$ (B) π (C) $-\pi/2$ (D) 0

Ans. D

Sol. -

Q10. (a)

List I

List

II

P. The number of polynomials $f(x)$ with non-negative integer coefficients 1. 8

of degree ≤ 2 , satisfying $f(0) = 0$ and $\int_0^1 f(x)dx = 1$, is

Q. The number of points in the interval $[-\sqrt{13}, \sqrt{13}]$ at which 2. 2

$f(x) = \sin(x^2) + \cos(x^2)$ attains its maximum value, is

R. $\int_{-2}^2 \frac{3x^2}{(1+e^x)} dx$ equals 3.

4

S. $\frac{\left(\int_{-\frac{1}{2}}^{\frac{1}{2}} \cos 2x \log \left(\frac{1+x}{1-x}\right) dx\right)}{\left(\int_0^{\frac{1}{2}} \cos 2x \log \left(\frac{1+x}{1-x}\right) dx\right)}$ equals

4. 0

P Q R S

(A) 3 2 4 1

(C) 3 2 1 4

P Q R S

(B) 2 3 4 1

(D) 2 3 1 4

(b) Let $f: (0, \infty) \rightarrow \mathbb{R}$ be given by $f(x) = \int_{\frac{1}{x}}^x e^{-(t+\frac{1}{t})} \frac{dt}{t}$. then

(A) $f(x)$ is monotonically increasing on $[1, \infty)$ (B) $f(x)$ is monotonically decreasing on $(0, 1)$

(C) $f(x) + f\left(\frac{1}{x}\right) = 0$, for all $x \in (0, \infty)$

(D) $f(2^x)$ is an odd function of x

on $\mathbb{R}$

Ans. (a) D, (b) ACD

Sol. -

Q11.

Let $f: (0, \infty) \rightarrow \mathbb{R}$ be given by $f(x) = \int_{\frac{1}{x}}^x e^{-(t+\frac{1}{t})} \frac{dt}{t}$. then :

(A) $f(x)$ is monotonically increasing on $[1, \infty)$

(B) $f(x)$ is monotonically decreasing on $(0, 1)$

(C) $f(x) + f\left(\frac{1}{x}\right) = 0$, for all $x \in (0, \infty)$ (D) $f(2^x)$ is an odd function of x on $\mathbb{R}$

Ans. A, C, D

Sol.

$$f(x) = \int_{\frac{1}{x}}^x e^{-\left(t + \frac{1}{t}\right)} dt$$

$$\Rightarrow f'(x) = \frac{e^{-(x+\frac{1}{x})}}{x} + \frac{1}{x^2} e^{-(-x+\frac{1}{x})} \cdot x$$

$$= e^{-(x+\frac{1}{x})} \cdot \frac{2}{x}$$

$$\Rightarrow f'(x) > 0 \forall x \in (0, \infty)$$

$$\Rightarrow f(x) \text{ is increasing in } (0, \infty)$$

$$\Rightarrow f(x) \text{ is increasing in } (0, \infty)$$

$$\text{Now } f(x) + f\left(\frac{1}{x}\right) = \int_{\frac{1}{x}}^x \frac{e^{-(t+\frac{1}{t})}}{t} dt + \int_x^{\frac{1}{x}} \frac{e^{-(t+\frac{1}{t})}}{t} dt$$

$$= \int_{\frac{1}{x}}^{\frac{1}{x}} \frac{e^{-(t+\frac{1}{t})}}{t} dt = 0$$

Similarly

$$f(2^x) + f\left(\frac{1}{2^x}\right) = 0$$

$$\Rightarrow f(2^{-x}) = -f(2^x)$$

Q12. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \begin{cases} [x], & x \leq 2 \\ 0, & x > 2 \end{cases}$

where $[x]$ is the greatest integer less than or equal to x .

If $I = \int_{-1}^2 \frac{xf(x^2)}{2+f(x+1)} dx$, then the value of $(4I - 1)$ is

Ans. 0

Sol. -

Q13. If $\alpha = \int_0^1 \left(e^{9x+3 \tan^{-1} x} \right) \left(\frac{12+9x^2}{1+x^2} \right) dx$ where $\tan^{-1} x$ takes only principal values, then the value of $\left(\log_e |1+\alpha| - \frac{3\pi}{4} \right)$ is :

Ans. 9

Sol. -

Q14. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous odd function, which vanishes exactly at one point and $f(1) = \frac{1}{2}$.

Suppose that $F(x) = \int_{-1}^x f(t)dt$ for all $x \in [-1, 2]$ and $G(x) = \int_{-1}^x f|f(f(t))|dt$ for all $x \in [-1, 2]$.

If $\lim_{x \rightarrow 1} \frac{f(x)}{G(x)} = \frac{1}{14}$, then the value of $f\left(\frac{1}{2}\right)$ is :

Ans. 7

Sol. -

Q15. The option (s) with the values of a and L that satisfy the following equation is (are)

$$\frac{\int_0^{4\pi} e^t (\sin^6 at + \cos^4 at) dt}{\int_0^{\pi} e^t (\sin^6 at + \cos^4 at) dt} = L ?$$

(A) $a = 2, L = \frac{e^{4\pi} - 1}{e^{\pi} - 1}$ (B) $a = 2, L = \frac{e^{4\pi} + 1}{e^{\pi} + 1}$ (C) $a = 4, L = \frac{e^{4\pi} - 1}{e^{\pi} - 1}$

(D) $a = 4, L = \frac{e^{4\pi} - 1}{e^{\pi} - 1}$

Ans. C, A

Sol. -

Q16. Let $f(x) = 7 \tan^8 x + 7 \tan^6 x - 3 \tan^4 x - 3 \tan^2 x$ for all $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then the correct expression (s) is (are) :

(A) $\int_0^{\pi/4} x f(x) dx = \frac{1}{12}$ (B) $\int_0^{\pi/4} f(x) dx = 0$ (C) $\int_0^{\pi/4} x f(x) dx = \frac{1}{6}$ (D) $\int_0^{\pi/4} f(x) dx = 1$

Ans. B, A

Sol. -

Q17. Let $f'(x) = \frac{192x^3}{2 + \sin^4 \pi x}$ for all $x \in \mathbb{R}$ with $f\left(\frac{1}{2}\right) = 0$. If $m \leq \int_{1/2}^1 f(x)dx \leq M$, then the possible values of m and M are :

- (A) $m = 13, M = 24$ (B) $m = \frac{1}{4}, M = \frac{1}{2}$ (C) $m = -11, M = 0$ (D) $m = 1, M = 12$

Ans. D

Sol. -

Paragraph

Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be a thrice differentiable function. Suppose that $F(1) = 0, F(3) = -4$ and $F'(x) < 0$ for all $x \in (1/2, 3)$. Let $f(x) = xF(x)$ for all $x \in \mathbb{R}$.

Q18. The correct statement (s) is (are)

- (A) $f'(1) < 0$ (B) $f(2) < 0$ (C) $f(x) \neq 0$ for any $x \in (1, 3)$
(D) $f(x) = 0$ for some $x \in (1, 3)$

Ans. 5f2a3157c2ce903f3b53628b, 5f2a3157c2ce903f3b53628a, 5f2a3157c2ce903f3b53628c

Sol. -

Paragraph

Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be a thrice differentiable function. Suppose that $F(1) = 0, F(3) = -4$ and $F'(x) < 0$ for all $x \in (1/2, 3)$. Let $f(x) = xF(x)$ for all $x \in \mathbb{R}$.

Q19. If $\int_1^3 x^2 F'(x) dx = -12$ and $\int_1^3 x^3 F''(x) dx = 40$, then the correct expression (s) is (are)

- (A) $9f(3) + f(1) - 32 = 0$ (B) $\int_1^3 f(x)dx = 12$ (C) $9f(3) - f(1) + 32 = 0$
(D) $\int_1^3 f(x)dx = -12$

Ans. 5f2a32c29cb8c03ef7299a78, 5f2a32c29cb8c03ef7299a79

Sol. -

Q20. The total number of distinct $x \in [0, 1]$ for which $\int_0^x \frac{t^2}{1+t^4} dt = 2x - 1$ is.

Ans. 1

Sol. -

Q21. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x^2 \cos x}{1 + e^x} dx$ is equal to

- (A) $\frac{\pi^2}{4} - 2$ (B) $\frac{\pi^2}{4} + 2$ (C) $\pi^2 - e^{\frac{\pi}{2}}$ (D) $\pi^2 + e^{\frac{\pi}{2}}$

Ans. A

Sol. -

Q22. Let $f(x) = \lim_{n \rightarrow \infty} \left(\frac{n^n (x+n)(x+\frac{n}{2}) \dots (x+\frac{n}{n})}{n!(n^2+n^2)(x^2+\frac{n^2}{4}) \dots (x^2+\frac{n^2}{n^2})} \right)^{\frac{x}{n}}$, for all $x > 0$. Then

- (A) $f\left(\frac{1}{2}\right) \geq f(1)$ (B) $f\left(\frac{1}{3}\right) \leq f\left(\frac{2}{3}\right)$ (C) $f(2) \leq 0$ (D) $\frac{f'(3)}{f'(2)} \geq \frac{f'(2)}{f'(3)}$

Ans. B, C

Sol. -

Q23. If $I = \sum_{k=1}^{98} \int_k^{k+1} \frac{k+1}{x(x+1)} dx$, then :

- (A) $I > \frac{49}{50}$ (B) $I < \frac{49}{50}$ (C) $I < \log_e 99$ (D) $I > \log_e 99$

Ans. C, A

Sol. -

Q24. If $g(x) = \int_{\sin x}^{\sin(2x)} \sin^{-1}(t) dt$, then :

- (A) $g'\left(-\frac{\pi}{2}\right) = -2\pi$ (B) $g'\left(-\frac{\pi}{2}\right) = 2\pi$ (C) $g'\left(\frac{\pi}{2}\right) = 2\pi$ (D) $g'\left(\frac{\pi}{2}\right) = -2\pi$

Ans. -

Sol. -

Q25. (a) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $f(0) = 0$, $f\left(\frac{\pi}{2}\right) = 3$ and $f'(0) = 1$.

If for $x \int_x^{\frac{\pi}{2}} [f(t) \operatorname{cosec} t - \cot t \operatorname{cosec} t f(t)] dt$ for $x \in (0, \frac{\pi}{2}]$, then $\lim_{n \rightarrow 0} g(x) =$

(b) Let $f: \square \rightarrow (0, 1)$ be a continuous function. Then, which of the following function(s) has(have) the value zero at some point in the interval $(0, 1)$?

- (A) $x^9 - f(x)$ (B) $f(x) + \int_0^{\frac{\pi}{2}} f(t) \sin t dt$ (C) $x - \int_0^{\frac{\pi}{2}-x} f(t) \cos t dt$ (D) $ex - \int_0^x f(t) \sin t dt$

Ans. (a) 2, (b) AC

Sol. -

Q26. The value of the integral $\int_0^{\frac{1}{2}} \frac{1 + \sqrt{3}}{((x+1)^2(1-x)^6)^{1/4}} dx$ is

Ans. 2

Sol. -

Q27. For each positive integer n , let $y_n = \frac{1}{n}((n+1)(n+2)\dots(n+n))^{\frac{1}{n}}$

For $x \in \mathbb{R}$, let $[x]$ be the greatest integer less than or equal to x . If $\lim_{n \rightarrow \infty} y_n = L$, then the value of $[L]$ is _____

Ans. 1

Sol. -

Q28. For $a \in \mathbb{R}$, $|a| > 1$, Let

$$\lim_{n \rightarrow \infty} \left(\frac{1 + \sqrt[3]{2} + \dots + \sqrt[3]{n}}{n^{7/3} \left(\frac{1}{(an+1)^2} + \frac{1}{(an+2)^2} + \dots + \frac{1}{(an+n)^2} \right)} \right) = 54$$

Then the possible value (s) of a is/are

(A) -6 (B) 8 (C) 7 (D) -9

Ans. D, B

Sol. -

Q29. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = (x-1)(x-2)(x-5)$ define

$$f(x) = \int_0^x f(t) dt, x > 0$$

Then which of the following options is/are correct

(A) $F(x) \neq 0$ for all $x \in (0,5)$ (B) F has a local maximum at $x = 2$

(C) F has a local minimum at $x = 1$

(D) F has two local maxima and one local minimum in $(0, \infty)$

Ans. C, B, A

Sol. -

Q30. If $I = \frac{2}{\pi} \int_{-\pi/4}^{\pi/4} \frac{dx}{(1 + e^{\sin x})(2 - \cos 2x)}$ then $27I^2$ equals

Ans. 4

Sol. -

Q31. समाकल $\int_0^{\pi/2} \frac{3\sqrt{\cos \theta}}{(\sqrt{\cos \theta} + \sqrt{\sin \theta})^5} d\theta$ का मान बराबर _____

Ans. 0.5

Sol. -

Q32. Which of the following inequalities is/are TRUE?

(A) $\int_0^1 x \cos x dx \geq \frac{3}{8}$ (B) $\int_0^1 x \sin x dx \geq \frac{3}{10}$ (C) $\int_0^1 x^2 \cos x dx \geq \frac{1}{2}$

(D) $\int_0^1 x^2 \sin x dx \geq \frac{2}{9}$

Ans. A, B, D

Sol. (A) $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$\cos x \geq 1 - \frac{x^2}{2}$$

$$\int_0^1 x \cos x \geq \int_0^1 x \left(1 - \frac{x^2}{2}\right) = \frac{1}{2} - \frac{1}{8}$$

$$\int_0^1 x \cos x \geq \frac{3}{8} \text{ (True)}$$

(B) $\sin x \geq x - \frac{x^3}{6}$

$$\int_0^1 x \sin x \geq \int_0^1 x \left(x - \frac{x^3}{6}\right) dx$$

$$\int_0^1 x \sin x \geq \frac{1}{3} - \frac{1}{30} \Rightarrow \int_0^1 x \sin x dx \geq \frac{3}{8}$$

(True)

(D) $\int_0^1 x^2 \sin x dx \geq \int_0^1 x^2 \left(x - \frac{x^3}{6}\right) dx$

$$\int_0^1 x^2 \sin x dx \geq \frac{1}{4} - \frac{1}{36}$$

$$\int_0^1 x^2 \sin x dx \geq \frac{2}{9} \text{ (True)}$$

(C) $\cos x < 1$

$$x^2 \cos x < x^2$$

$$\int_0^1 x^2 \cos x dx < \int_0^1 x^2 dx$$

$$\int_0^1 x^2 \cos x dx < \frac{1}{3}$$

So option 'C' is incorrect.

Q33. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that its derivative f' is continuous and $f(\pi) = -6$.

If $F: [0, \pi] \rightarrow \mathbb{R}$ is defined by $F(x) = \int_0^x f(t) dt$, and if $\int_0^\pi (f(x) + F(x)) \cos x dx = 2$

then the value of $f(0)$ is _____

Ans. 4

Sol.

$$F(x) = \int_0^x f(t) dt$$

differentiating wrt x

$$F'(x) = f(x) \dots (1)$$

$$\int_0^\pi (f'(x) \cos x + F(x) \cos x) dx = 2$$

$$\Rightarrow \int_0^\pi f'(x) \cos x dx + \int_0^\pi F(x) \cos x dx = 2$$

Using integration by parts

$$\left[f(x) \cos x \right]_0^\pi + \int_0^\pi \sin x \cdot f(x) dx + \left[F(x) \cdot \sin x \right]_0^\pi - \int_0^\pi \sin x F'(x) dx = 2$$

Using (1)

$$\left[f(x) \cos x \right]_0^\pi + \int_0^\pi \sin x \cdot f(x) dx + \left[F(x) \cdot \sin x \right]_0^\pi - \int_0^\pi \sin x f(x) dx = 2$$

$$\Rightarrow -f(\pi) - f(0) + F(\pi) \cdot 0 - F(0) \cdot 0 = 2$$

$$\Rightarrow -f(\pi) - f(0) = 2$$

$$\Rightarrow 6 - f(0) = 2$$

$$\Rightarrow f(0) = 4$$

Q34. Let $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow \mathbb{R}$ be a continuous function such that $f(0) = 1$ and $\int_0^{\frac{\pi}{3}} f(t) dt = 0$.

Then which of the following statements is (are) TRUE ?

(A) The equation $f(x) - 3 \cos 3x = 0$ has at least one solution in $\left(0, \frac{\pi}{3}\right)$

(B) The equation $f(x) - 3 \sin 3x = -\frac{6}{\pi}$ has at least one solution in $\left(0, \frac{\pi}{3}\right)$

$$(C) \lim_{x \rightarrow 0} \frac{x \int_0^x f(t) dt}{1 - e^{x^2}} = -1 \quad (D) \lim_{x \rightarrow 0} \frac{\sin x \int_0^x f(t) dt}{x^2} = -1$$

Ans. A, B, C

Sol. $\int_0^{\frac{\pi}{3}} (f(x) - 3 \cos 3x) dx = \int_0^{\frac{\pi}{3}} f(x) dx - \int_0^{\frac{\pi}{3}} 3 \cos 3x dx = 0 - \sin 3x \Big|_0^{\pi/3} = 0$

$$\Rightarrow f(x) - 3 \cos 3x = 0 \text{ has atleast one solution in } \left(0, \frac{\pi}{3}\right)$$

$$(B) \int_0^{\frac{\pi}{3}} \left(f(x) - 3 \sin 3x + \frac{6}{\pi} \right) dx = 3x dx + \int_0^{\frac{\pi}{3}} \frac{6}{\pi} dx = 0 + \cos 3x \Big|_0^{\pi/3} + \frac{6}{\pi} \cdot \frac{\pi}{3} = 0$$

$$\Rightarrow f(x) - 3 \sin 3x + \frac{6}{\pi} = 0 \text{ has atleast one solution in } \left(0, \frac{\pi}{3}\right)$$

$$(C) \lim_{x \rightarrow \infty} \frac{x \int_0^x f(t) dt}{1 - e^{x^2}} \left(\frac{0}{0} \right)$$

Use L'hospital Rule

$$\lim_{x \rightarrow 0} \frac{x \cdot f(x) + \int_0^x f(t) dt}{-2xe^{x^2}} \left(\frac{0}{0} \right)$$

Use L'Hospital Rule again

$$\lim_{x \rightarrow 0} \frac{x \cdot f'(x) + 2f(x)}{-2e^{x^2} - 4x^2 \cdot e^{x^2}} = \frac{2 \cdot f(0)}{-2} = -1$$

$$(D) \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) \cdot \frac{\int_0^x f(t) dt}{x}$$

$$\lim_{x \rightarrow 0} \frac{\int_0^x f(t) dt}{x} \left(\frac{0}{0} \right)$$

Use L'Hospital Rule

$$\lim_{x \rightarrow 0} \frac{f(x)}{1} = f(0) = 1$$

As ABC

Q35. Let $g_i : \left[\frac{\pi}{8}, \frac{3\pi}{8} \right] \rightarrow R, i = 1, 2$, and $f : \left[\frac{\pi}{8}, \frac{3\pi}{8} \right] \rightarrow R$ be functions such that

$$g_1(x) = 1, g_2(x) = |4x - \pi|$$

$$\text{and } f(x) = \sin^2 x, \text{ for all } x \in \left[\frac{\pi}{8}, \frac{3\pi}{8} \right]$$

$$\text{Define } S_i = \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} f(x) \cdot g_i(x) dx, \quad i = 1, 2$$

The value of $\frac{16S_1}{\pi}$ is ____.

Ans. 1.99, 2.01

Sol.

$$S_1 = \int_{\pi/8}^{3\pi/8} f(x) \cdot 1 dx = \int_{\pi/8}^{3\pi/8} \sin^2 x dx \quad \dots\dots\dots(1)$$

$$\text{Apply } \int_a^b f(a+b-x) dx = \int_a^b f(x) dx$$

$$S_1 = \int_{\pi/8}^{3\pi/8} \cos^2 x dx \quad \dots\dots\dots(2)$$

$$(1) + (2) \Rightarrow 2S_1 = \int_{\pi/8}^{3\pi/8} dx = \frac{\pi}{4} \Rightarrow S_1 = \frac{\pi}{8}$$

$$\frac{16S_1}{\pi} = 2$$

$$S_2 = \int_{\pi/8}^{3\pi/8} \sin^2 x |4x - \pi| dx \quad \dots\dots\dots(3)$$

$$\text{Apply } \int_a^b f(a+b-x) dx = \int_a^b f(x) dx$$

$$S_2 = \int_{\pi/8}^{3\pi/8} \cos^2 x |4x - \pi| dx \quad \dots\dots\dots(4)$$

$$(1) + (2) \Rightarrow 2S_2 = \int_{\pi/8}^{3\pi/8} |4x - \pi| dx = \frac{\pi^2}{16}$$

$$A_1 + A_2 = \frac{1}{2} \times \frac{\pi}{4} \times \frac{\pi}{2} = \frac{\pi^2}{16}$$

$$\Rightarrow S_2 = \frac{\pi^2}{32}$$

$$\Rightarrow \frac{48}{\pi^2} S_2 = 1.5$$

Q36. Let $g_i : \left[\frac{\pi}{8}, \frac{3\pi}{8}\right] \rightarrow R, i = 1, 2$, and $f : \left[\frac{\pi}{8}, \frac{3\pi}{8}\right] \rightarrow R$ be functions such that

$$g_1(x) = 1, g_2(x) = |4x - \pi|$$

$$\text{and } f(x) = \sin^2 x, \text{ for all } x \in \left[\frac{\pi}{8}, \frac{3\pi}{8}\right]$$

The value of $\frac{48S_2}{\pi^2}$ is ____.

Ans. 1.49, 1.51

Sol.

$$S_1 = \int_{\pi/8}^{3\pi/8} f(x) \cdot 1 dx = \int_{\pi/8}^{3\pi/8} \sin^2 x dx \quad \dots\dots\dots(1)$$

$$\text{Apply } \int_a^b f(a+b-x) dx = \int_a^b f(x) dx$$

$$S_1 = \int_{\pi/8}^{3\pi/8} \cos^2 x dx \quad \dots\dots\dots(2)$$

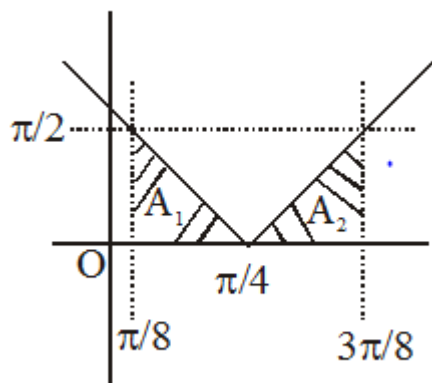
$$(1) + (2) \Rightarrow 2S_1 = \int_{\pi/8}^{3\pi/8} dx = \frac{\pi}{4} \Rightarrow S_1 = \frac{\pi}{8}$$

$$\frac{16S_1}{\pi} = 2$$

$$S_2 = \int_{\pi/8}^{3\pi/8} \sin^2 x |4x - \pi| dx \quad \dots\dots\dots(3)$$

$$\text{Apply } \int_a^b f(a+b-x)dx = \int_a^b f(x)dx$$

$$S_2 = \int_{\pi/8}^{3\pi/8} \cos^2 x |4x - \pi| dx \quad \dots\dots\dots(4)$$



$$(1) + (2) \Rightarrow 2S_2 = \int_{\pi/8}^{3\pi/8} |4x - \pi| dx = \frac{\pi^2}{16}$$

$$A_1 + A_2 = \frac{1}{2} \times \frac{\pi}{4} \times \frac{\pi}{2} = \frac{\pi^2}{16}$$

$$\Rightarrow S_2 = \frac{\pi^2}{32}$$

$$\Rightarrow \frac{48}{\pi^2} S_2 = 1.5$$

Paragraph

Let $\psi_1 : [0, \infty) \rightarrow R$, $\psi_2 : [0, \infty) \rightarrow R$, $f : [0, \infty) \rightarrow R$ and $g : [0, \infty) \rightarrow R$ be functions such that

$$f(0) = g(0) = 0,$$

$$\psi_1(x) = e^{-x} + x, \quad x \geq 0,$$

$$\psi_2(x) = x^2 - 2x - 2e^{-x} + 2, \quad x \geq 0,$$

$$f(x) = \int_{-x}^x (|t| - t^2)e^{-t^2} dt, \quad x > 0$$

$$\text{and } g(x) = \int_0^{x^2} \sqrt{t}e^{-t} dt, \quad x > 0.$$

Q37. Which of the following statements is TRUE ?

(A) $\psi_1(x) \leq 1$, for all $x > 0$ (B) $\psi_2(x) \leq 0$, for all $x > 0$

(C) $f(x) \geq 1 - e^{-x^2} - \frac{2}{3}x^3 + \frac{2}{5}x^5$, for all $x \in \left(0, \frac{1}{2}\right)$

(D) $g(x) \leq \frac{2}{3}x^3 - \frac{2}{5}x^5 + \frac{1}{7}x^7$, for all $x \in \left(0, \frac{1}{2}\right)$

Ans. D

Sol. $\psi_1(x) = e^{-x} + x$

$$\psi_1'(x) = -e^{-x} + 1$$

$$\psi_1'(x) > 0$$

hence $\psi_1(x)$ is increasing then

$$\psi_1(x) \geq \psi_1(0)$$

$$\psi_1(x) \geq 1$$

So option (A) is wrong.

Option (B)

$$\psi_2(x) = x^2 - 2x - 2e^{-x} + 2$$

$$\psi_2'(x) = 2x - 2 + 2e^{-x}$$

$$\psi_2'(x) > 0$$

hence $\psi_2'(x)$ is increasing.

then

$$\psi_2(x) \geq \psi_2(0)$$

$$\psi_2(x) \geq 0$$

hence option (B) is wrong.

Option (C)

$$f(x) \geq 1 - e^{-x^2} - \frac{2}{3}x^3 + \frac{2}{5}x^5$$

Consider

$$H(x) = f(x) - 1 + e^{-x^2} + \frac{2}{3}x^3 - \frac{2}{5}x^5$$

$$H'(x) = f'(x) - 2xe^{-x^2} + 2x^2 - 2x^4$$

$$\text{but } f'(x) = 2(x - x^2)e^{-x^2}$$

Now

$$H'(x) = 2x^2(1 - x^2 - e^{-x^2})$$

$$H'(x) = 2x^2(1 - x^2 - 1 + \frac{x^2}{1} - \frac{x^4}{2} + \dots) < 0$$

then $H(x)$ is decreasing

$$H(x) \leq H(0)$$

$$H(x) \leq f(0)$$

$$H(x) \leq 0$$

hence

$$f(x) \geq 1 - e^{-x^2} - \frac{2}{3}x^3 + \frac{2}{5}x^5$$

hence option (C) is wrong.

Option (D)

$$g(x) \leq \frac{2}{3}x^3 - \frac{2}{5}x^5 + \frac{1}{7}x^7$$

Let consider

$$H(x) = g(x) - \frac{2}{3}x^3 + \frac{2}{5}x^5 - \frac{1}{7}x^7$$

$$H'(x) = g'(x) - 2x^2 + 2x^4 - x^6$$

$$\text{but } g'(x) = 2x^2 e^{-x^2}$$

$$H'(x) = 2x^2 e^{-x^2} - 2x^2 + 2x^4 - x^6$$

$$H'(x) = 2x^2 \left(1 - \frac{x^2}{1} + \frac{x^4}{2} - \frac{x^6}{3} \dots \right) - 2x^2 + 2x^4 - x^6 < 0$$

$H(x)$ is decreasing.

So

$$H(x) \leq H(0)$$

$$g(x) - \frac{2}{3}x^3 + \frac{2}{5}x^5 - \frac{1}{7}x^7 \leq 0$$

$$g(x) \leq \frac{2}{3}x^3 - \frac{2}{5}x^5 + \frac{1}{7}x^7$$

hence option (D) is correct.

Paragraph

Let $\psi_1 : [0, \infty) \rightarrow R$, $\psi_2 : [0, \infty) \rightarrow R$, $f : [0, \infty) \rightarrow R$ and $g : [0, \infty) \rightarrow R$ be functions such that $f(0) = g(0) = 0$,

$$\psi_1(x) = e^{-x} + x, \quad x \geq 0,$$

$$\psi_2(x) = x^2 - 2x - 2e^{-x} + 2, \quad x \geq 0,$$

$$f(x) = \int_{-x}^x (|t| - t^2) e^{-t^2} dt, \quad x > 0$$

$$\text{and } g(x) = \int_0^{x^2} \sqrt{t} e^{-t} dt, \quad x > 0.$$

Q38. Which of the following statements is TRUE ?

$$(A) \quad f(\sqrt{\ln 3}) + g(\sqrt{\ln 3}) = \frac{1}{3}$$

(B) For every $x > 1$, there exists an $\alpha \in (1, x)$ such that $\psi_1(x) = 1 + \alpha x$

(C) For every $x > 0$, there exists a $\beta \in (0, x)$ such that $\psi_2(x) = 2x(\psi_1(\beta) - 1)$

(D) f is an increasing function on the interval $\left[0, \frac{3}{2}\right]$

Ans. C

Sol.

$$f(x) = 2 \int_0^x (t - t^2) e^{-t^2} dt$$

$$f(x) = 2 (x - x^2) e^{-x^2}$$

$$g'(x) = 2x^2 e^{-x^2}$$

Option (A)

Let consider

$$f(x) + g'(x) = 2 + (x - x^2)e^{-x^2} + 2x^2 e^{-x^2}$$

Integrating both side

$$f(x) + g(x) = -e^{-x^2} + c$$

put $x = 0$

then $c = 1$

$$f(x) + g(x) = 1 - e^{-x^2}$$

$$f(\sqrt{\ln 3}) + g(\sqrt{\ln 3}) = 1 - e^{-\ln^2 3} = 1 - \frac{1}{3} = \frac{2}{3}$$

hence option (A) is wrong.

Option (B)

Let consider

$$H(x) = \psi_1(x) - 1 - \alpha x$$

$$H(1) = \psi_1(1) - 1 - \alpha$$

$$H(1) = e^{-1} + 1 - 1 - \alpha$$

$$H(1) = e^{-1} - \alpha < 0$$

$$H'(x) = \psi_1'(x) - \alpha$$

$$H'(x) = -e^{-x} + 1 - \alpha < 0$$

hence $H(x)$ is decreasing.

So $H(x)$ is always negative and $H(x) \neq 0$

Hence option (B) is wrong.

Option (C)

$$\psi_2(x) = x^2 - 2x - 2e^{-x} + 2$$

Applying LMVT in $(0, x)$

$$\psi_2'(\beta) = \frac{\psi_2(x) - \psi_2(0)}{x - 0}$$

$$\psi_2'(\beta) = \frac{\psi_2(x) - 0}{x} \dots\dots\dots(1)$$

$$\psi_2'(x) = 2x - 2 + 2e^{-x}$$

$$\psi_2'(\beta) = 2\beta - 2 + 2e^{-\beta}$$

$$\psi_2'(\beta) = 2(\beta + e^{-\beta}) - 2$$

$$\psi_2'(\beta) = 2\psi_1(\beta) - 2$$

Now by (1)

$$2(\psi_1(\beta) - 1) = \frac{\psi_2(x)}{x}$$

$$\psi_2(x) = 2x(\psi_1(\beta) - 1)$$

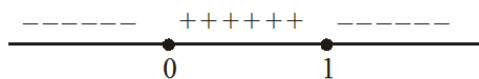
hence option (C) is correct.

Option (D)

$$\int_{-x}^x (|t| - t^2)e^{-t^2} dt$$

$$f(x) = 2(x - x^2)e^{-x^2}$$

$$f'(x) = 2x(1 - x)e^{-x^2}$$



Q39. For any real number x , let $[x]$ denote the largest integer less than or equal to x . If

$$I = \int_0^{10} \left[\sqrt{\frac{10x}{x+1}} \right] dx, \text{ then the value of } 9I \text{ is } \underline{\hspace{2cm}}.$$

Ans. 182

Sol. $f(x) = \frac{10x}{x+1}$
 $f'(x) = \frac{10x}{(x+1)^2} > 0$

hence $f'(x)$ is increasing

Now

$$\int_0^{\frac{1}{9}} 0dx + \int_{\frac{1}{9}}^{\frac{2}{3}} 1dx + \int_{\frac{2}{3}}^9 2dx + \int_9^{10} 3dx = \frac{182}{9}$$

$$\text{Now } 9I = 182$$

$$\text{Ans} = 182$$

Q40. Consider the equation $\int_1^e \frac{(\log_e x)^{1/2}}{x(a - (\log_e x)^{3/2})^2} dx = 1, a \in (-\infty, 0) \cup (1, \infty)$. Which of the following statements is/are TRUE ?

- (A) No a satisfies the above equation (B) An integer a satisfies the above equation
 (C) An irrational number a satisfies the above equation
 (D) More than one a satisfy the above equation

Ans. C, D

Sol. Let $I = \int_1^e \frac{(\ln x)^{1/2} dx}{x(a - (\ln x)^{3/2})^2}$

$$\text{put } a - (\ln x)^{3/2} = t$$

$$\Rightarrow -\frac{3}{2}(\ln x)^{1/2} \cdot \frac{1}{x} dx = dt$$

$$\therefore I = \int_a^{a-1} \frac{\left(-\frac{2}{3}\right) dt}{t^2}$$

$$= \left(-\frac{2}{3} \right) \frac{t^{-2+1}}{-2+1} \Big|_a^{a-1}$$

$$= \frac{2}{3t} \Big|_a^{a-1} = \frac{2}{3} \left(\frac{1}{a-1} - \frac{1}{a} \right)$$

$$\begin{aligned}\therefore I &= \left(\frac{2}{3}\right) \frac{1}{a(a-1)} = 1 \\ \Rightarrow 2 &= 3a^2 - 3a \\ \Rightarrow 3a^2 - 3a - 2 &= 0 \\ \Rightarrow a &= \frac{3 \pm \sqrt{9 - 4(3)(-2)}}{6} \\ a &= \frac{3 + \sqrt{33}}{6}, \frac{3 - \sqrt{33}}{6}\end{aligned}$$

Q41. The greatest integer less than or equal to $\int_1^2 \log_2(x^3 + 1) dx + \int_1^{\log_2 9} (2^x - 1)^{\frac{1}{3}} dx$ is _____.

Ans. 5

Sol. $y = \int_1^2 \log_2(x^3 + 1) dx + \int_1^{2\log_2 3} (2^x - 1)^{\frac{1}{3}} dx$
 $= 4\log_2 3 - 1$
 we know $\int_a^b f(x) dx + \int_{f(a)}^{f(b)} (f^{-1}(x)) dx = bf(b) - af(a)$
 So, $[y] = 5$

Q42. Let $f : (0, 1) \rightarrow R$ be the function defined as $f(x) = \sqrt{n}$ if $x \in \left[\frac{1}{n+1}, \frac{1}{n}\right)$ where $n \in N$.

Let $g : (0, 1) \rightarrow R$ be a function such that $\int_{x^2}^x \sqrt{\frac{1-t}{t}} dt < g(x) < 2\sqrt{x}$ for all $x \in (0, 1)$.

Then $\lim_{x \rightarrow 0} f(x)g(x)$

(A) does NOT exist (B) is equal to 1 (C) is equal to 2 (D) is equal to 3

Ans. C

Sol. $f(x) = \sqrt{x}$
 $\frac{1}{n+1} \leq x < \frac{1}{n}$
 $n < \frac{1}{x} \leq n+1$
 $n < \frac{1}{x}$
 $n \geq \frac{1}{x} - 1$
 $n \geq \frac{1-x}{x}$
 $\frac{1-x}{x} \leq n < \frac{1}{x}$
 $\sqrt{\frac{1-x}{x}} \leq \sqrt{n} < \frac{1}{\sqrt{x}}$
 $\sqrt{\frac{1-x}{x}} \leq f(x) < \frac{1}{\sqrt{x}}$

$$\int_{x^2}^x \sqrt{\frac{1-t}{t}} dt \times \sqrt{\frac{1-x}{x}} \leq f(x)g(x) < \frac{1}{\sqrt{x}} \times 2\sqrt{x}$$

$$\lim_{x \rightarrow 0} \int_{x^2}^x \sqrt{\frac{1-t}{t}} dt \sqrt{\frac{1-x}{x}} \leq \lim_{x \rightarrow 0} f(x)g(x) < \lim_{x \rightarrow 0} 2$$

$$\lim_{x \rightarrow 0} \frac{\int_{x^2}^x \left(\sqrt{\frac{1-t}{t}} \right) dt}{\sqrt{\frac{x}{1-x}}} \left(\frac{0}{0} \right)$$

L.H. Rule

$$\lim_{x \rightarrow 0} \frac{\sqrt{\frac{1-x}{x}} - \sqrt{\frac{1-x^2}{x^2}} \times (2x)}{\frac{1}{2\sqrt{\frac{x}{1-x}}} \times \left(\frac{1-x+x}{(1-x)^2} \right)} = \lim_{x \rightarrow 0} \frac{2\sqrt{\frac{1-x}{x}} \left(1 - \sqrt{\frac{1+x}{x}} \times 2x \right)}{\sqrt{\frac{1-x}{x}} \times \frac{1}{(1-x)^2}} = 2$$

- Q43. Let $f: [1, \infty) \rightarrow \mathbb{R}$ be a differentiable function such that $f(1) = \frac{1}{3}$ and $3 \int_1^x f(t) dt = xf(x) - \frac{x^3}{3}, x \in [1, \infty)$. Let e denote the base of the natural logarithm. Then the value of $f(e)$ is

(A) $\frac{e^2 + 4}{3}$ (B) $\frac{\log_e 4 + e}{3}$ (C) $\frac{4e^2}{3}$ (D) $\frac{e^2 - 4}{3}$

Ans. C

Sol. Diff. w.r.t 'x'

$$3f(x) = f(x) + xf'(x) - x^2$$

$$\frac{dy}{dx} - \left(\frac{2}{x} \right) y = x$$

$$\text{IF} = e^{-2\ln x} = \frac{1}{x^2}$$

$$y \left(\frac{1}{x^2} \right) = \int x \cdot \frac{1}{x^2} dx$$

$$y = x^2 \ln x + cx^2$$

$$\therefore y(1) = \frac{1}{3} \Rightarrow c = \frac{1}{3}$$

$$y(e) = \frac{4e^2}{3}$$

- Q44. For $x \in \mathbb{R}$, let $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$. Then the minimum value of the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \int_0^{x \tan^{-1} x} \frac{e^{(t - \cos t)}}{1 + t^{2023}} dt$ is

Ans. 0

Sol.

$$f'(x) = \frac{\left(e^{x \tan^{-1} x - \cos(x \tan^{-1} x)}\right)}{1 + (x \tan^{-1} x)^{2023}} \left(\tan^{-1} x + \frac{x}{1 + x^2}\right)$$

$$f'(x) = 0$$

$$\Rightarrow \tan^{-1} x + \frac{x}{1 + x^2} = 0 \Rightarrow x = 0$$

$$\text{and } f'(0^+) > 0$$

$$f'(0^-) < 0$$

So min at $x = 0$

$$f(0) = 0$$

$$\text{Ans.} = 0$$

Q45.

Ans. -

Sol. -

Q46. **PARAGRAPH :**

Let $f : \left[0, \frac{\pi}{2}\right] \rightarrow [0, 1]$ be the function defined by $f(x) = \sin^2 x$ and let $g : \left[0, \frac{\pi}{2}\right] \rightarrow [0, \infty)$

be the function defined by $g(x) = \sqrt{\frac{\pi x}{2} - x^2}$.

(There are two questions based on PARAGRAPH "II")

The value of $2 \int_0^{\frac{\pi}{2}} f(x)g(x)dx - \int_0^{\frac{\pi}{2}} g(x)dx$ is _____.

Ans. 0

Sol.

$$\text{Let } I_1 = 2 \int_0^{\frac{\pi}{2}} \sin^2 x \sqrt{\frac{\pi x}{2} - x^2} dx \quad \text{_____ (1)}$$

King

$$I_1 = 2 \int_0^{\frac{\pi}{2}} \cos^2 x \sqrt{\frac{\pi x}{2} - x^2} dx \quad \text{_____ (2)}$$

(1) + (2)

$$2I_1 = 2 \int_0^{\frac{\pi}{2}} \sqrt{\frac{\pi x}{2} - x^2} dx$$

$$I_1 = \int_0^{\frac{\pi}{2}} \sqrt{\frac{\pi x}{2} - x^2} dx$$

$$\text{Let } I_2 = \int_0^{\frac{\pi}{2}} \sqrt{\frac{\pi x}{2} - x^2} dx$$

$$\begin{aligned}\text{Ans.} &= I_1 - I_2 \\ &= 0\end{aligned}$$

Q47. PARAGRAPH :

Let $f : \left[0, \frac{\pi}{2}\right] \rightarrow [0, 1]$ be the function defined by $f(x) = \sin^2 x$ and let $g : \left[0, \frac{\pi}{2}\right] \rightarrow [0, \infty)$ be the function defined by $g(x) = \sqrt{\frac{\pi x}{2} - x^2}$.

(There are two questions based on PARAGRAPH "II")

The value of $\frac{16}{\pi^3} \int_0^{\frac{\pi}{2}} f(x)g(x)dx$ is _____.

Ans. 0.25

Sol.

$$I = \frac{16}{\pi^3} \int_0^{\frac{\pi}{2}} \sin^2 x \sqrt{\frac{\pi x}{2} - x^2} dx \quad \text{_____ (1)}$$

Apply king property

$$I = \frac{16}{\pi^3} \int_0^{\frac{\pi}{2}} \cos^2 x \sqrt{\frac{\pi x}{2} - x^2} dx \quad \text{_____ (2)}$$

(1) + (2)

$$2I = \frac{16}{\pi^3} \int_0^{\frac{\pi}{2}} \sqrt{\frac{\pi x}{2} - x^2} dx$$

$$I = \frac{8}{\pi^3} \int_0^{\frac{\pi}{2}} \sqrt{\frac{\pi^2}{16} - \left(x - \frac{\pi}{4}\right)^2} dx$$

Put $x - \frac{\pi}{4} = t$

$$dx = dt$$

$$I = \frac{8}{\pi^3} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sqrt{\frac{\pi^2}{16} - t^2} dt$$

Even function

$$\begin{aligned} I &= \frac{16}{\pi^3} \int_0^{\frac{\pi}{4}} \sqrt{\frac{\pi^2}{16} - t^2} dt \\ &= \frac{16}{\pi^3} \left[t \sqrt{\frac{\pi^2}{16} - t^2} + \frac{\pi^2}{32} \sin^{-1} \left(\frac{t}{\frac{\pi}{4}} \right) \right]_0^{\frac{\pi}{4}} \\ &= \frac{16}{\pi^3} \times \frac{\pi^2}{32} \times \frac{\pi}{2} = \frac{1}{4} = 0.25 \end{aligned}$$

Exercise C1

Q1. If $\int \frac{\cos x - \sin x + 1 - x}{e^x + \sin x + x} dx = \ln + (f(x)) + g(x) + C$ where C is the constant of integration and

$f(x) + g(x)$ has the value equal to –

- (A) $e^x + \sin x + 2x$ (B) $e^x + \sin x$ (C) $e^x - \sin x$ (D) $e^x + \sin x + x$

Ans. B

Sol. -

Q2. $\int (\sin(101x) \cdot \sin^{99} x) dx$ equals

- (A) $\frac{\sin(100x)(\sin x)^{100}}{100} + C$ (B) $\frac{\cos(100x)(\sin x)^{100}}{100} + C$ (C) $\frac{\cos(100x)(\cos x)^{100}}{100} + C$
(D) $\frac{\sin(100x)(\sin x)^{100}}{101} + C$

Ans. A

Sol. -

Q3. $\int a^x \left(\ln x + \ln a \cdot \ln \left(\frac{x}{e} \right)^x \right) dx =$

- (A) $a^x \ln \left(\frac{e}{x} \right)^{2x} + C$ (B) $a^x \ln \left(\frac{x}{e} \right)^x + C$ (C) $a^x + \ln \left(\frac{x}{e} \right)^x + C$ (D) None of these

Ans. B

Sol. -

Q4. $\int \frac{x^2 - 1}{x^3 \sqrt{2x^4 - 2x^2 + 1}} dx$ is equal to

- (A) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x^2} + C$ (B) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x^3} + C$ (C) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x} + C$
(D) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{2x^2} + C$

Ans. D

Sol. -

Q5. $\int \frac{8x^{43} + 13x^{38}}{(x^{13} + x^5 + 1)^4} dx =$

- (A) $\frac{x^{39}}{3(x^{13} + x^5 + 1)^3} + C$ (B) $\frac{x^{39}}{(x^{13} + x^5 + 1)^3} + C$ (C) $\frac{x^{39}}{5(x^{13} + x^5 + 1)^5} + C$
(D) none of these

Ans. A

Sol. -

Q6. $\int_0^1 \frac{\tan^{-1} x}{x} dx =$

(A) $\int_0^{\pi/4} \frac{\sin x}{x} dx$ (B) $\int_0^{\pi/2} \frac{x}{\sin x} dx$ (C) $\frac{1}{2} \int_0^{\pi/2} \frac{x}{\sin x} dx$ (D) $\frac{1}{2} \int_0^{\pi/4} \frac{x}{\sin x} dx$

Ans. C

Sol. -

Q7. The value of the definite integral $\int_0^{\pi/2} (\sin x + \cos x) \cdot \sqrt{\frac{e^x}{\sin x}} dx$ equals

(A) $2\sqrt{e^{\pi/2}}$ (B) $\sqrt{e^{\pi/2}}$ (C) $2\sqrt{e^{\pi/2}} \cdot \cos 1$ (D) $\frac{1}{2}e^{\pi/4}$

Ans. A

Sol. -

Q8. The absolute value of $\frac{\int_0^{\pi/2} (x \cos x + 1)e^{\sin x} dx}{\int_0^{\pi/2} (x \sin x - 1)e^{\cos x} dx}$ is equal to

(A) e (B) πe (C) e/2 (D) π/e

Ans. A

Sol. -

Q9. If $\int_0^1 \frac{e^t dt}{1+t} = A$ then the value of $\int_0^1 \frac{e^t dt}{(1+t)^2}$ is :

(A) $A + \frac{e}{2} - 1$ (B) $A - \frac{e}{2} + 1$ (C) $A - \frac{e}{2} - 1$ (D) $A + \frac{e}{2} + 1$

Ans. B

Sol. -

Q10. If $\beta + 2 \int_0^1 x^2 e^{-x^2} dx = \int_0^1 e^{-x^2} dx$ then the value of β is

(A) e^{-1} (B) e (C) $1/2e$ (D) can not be determined

Ans. A

Sol. -

Q11. The value of the definite integral $\int_0^{\pi/2} \sin |2x - \alpha| dx$ where $\alpha \in [0, \pi]$

- (A) 1 (B) $\cos \alpha$ (C) $\frac{1 + \cos \alpha}{2}$ (D) $(1 + \sin \alpha)$

Ans. A

Sol. -

Q12. The value of the definite integral $\int_2^4 (x(3-x)(4+x)(6-x)(10-x) + \sin x) dx$ equals

- (A) $\cos 2 + \cos 4$ (B) $\cos 2 - \cos 4$ (C) $\sin 2 + \sin 4$ (D) $\sin 2 - \sin 4$

Ans. B

Sol. -

Q13. $\int_{-2}^0 (x^3 + 3x^2 + 3x + 3 + (x+1) \cos(x+1)) dx$ is equal to

- (A) -4 (B) 0 (C) 4 (D) 6

Ans. C

Sol. -

Q14. Value of the definite integral $\int_{-1/2}^{1/2} (\sin^{-1}(3x - 4x^3) - \cos^{-1}(4x^3 - 3x)) dx$

- (A) 0 (B) $-\frac{\pi}{2}$ (C) $\frac{7\pi}{2}$ (D) $\frac{\pi}{2}$

Ans. B

Sol. -

Q15. The value of the definite integral $\int_0^{\pi/2} \sqrt{\tan x} dx$, is

- (A) $\sqrt{2}\pi$ (B) $\frac{\pi}{\sqrt{2}}$ (C) $2\sqrt{2}\pi$ (D) $\frac{\pi}{2\sqrt{2}}$

Ans. B

Sol. -

- Q16. If $y(t) = \int_{-\ln t}^{\ln t} \cos(te^x) dx$, then $y'(1)$ equals
- (A) $2 \cos 1$ (B) $-2 \cos 1$ (C) $-\cos 1$ (D) $\cos 1$

Ans. A

Sol. -

- Q17. $\lim_{x \rightarrow \infty} \left(x^3 \int_{-1/x}^{1/x} \frac{\ln(1+t^2)}{1+e^t} dt \right)$ equals

(A) $1/3$ (B) $2/3$ (C) 1 (D) 0

Ans. A

Sol. -

- Q18. Let $C_n = \int_{\frac{1}{n+1}}^{\frac{1}{n}} \frac{\tan^{-1}(nx)}{\sin^{-1}(nx)} dx$ then $\lim_{n \rightarrow \infty} n^2 \cdot C_n$ equals

(A) 1 (B) 0 (C) -1 (D) $\frac{1}{2}$

Ans. D

Sol. -

- Q19. $\lim_{\lambda \rightarrow 0} \left(\int_0^1 (1+x)^\lambda dx \right)^{1/\lambda}$ is equal to

(A) $2 \ln 2$ (B) $\frac{4}{e}$ (C) $\ln \frac{4}{e}$ (D) 4

Ans. B

Sol. -

- Q20. For positive integers $k = 1, 2, 3, \dots, n$, let S_k denotes the area of ΔAOB_k (where 'O' is origin) such that $\angle AOB_k = \frac{k\pi}{2n}$, $OA = 1$ and $OB_k = k$. The value of the

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n S_k \text{ is}$$

(A) $\frac{2}{\pi^2}$ (B) $\frac{4}{\pi^2}$ (C) $\frac{8}{\pi^2}$ (D) $\frac{1}{2\pi^2}$

Ans. A

Sol. -

Q21. The interval $[0, 4]$ is divided into n equal sub-intervals by the points $x_0, x_1, x_2, \dots, x_{n-1}, x_n$ where $0 = x_0 < x_1 < x_2 < x_3 < \dots < x_n = 4$. If $\Delta x = x_i - x_{i-1}$ for $i = 1, 2, 3, \dots, n$ then $\lim_{\Delta x \rightarrow 0} \sum_{i=1}^n x_i \Delta x$ is equal to

- (A) 4 (B) 8 (C) $\frac{32}{3}$ (D) 16

Ans. B

Sol. -

Q22. If $f(\theta) = \frac{4}{3}(1 - \cos^6 \theta - \sin^6 \theta)$, then

$$\lim_{n \rightarrow \infty} \frac{1}{n} \left[\sqrt{f\left(\frac{1}{n}\right)} + \sqrt{f\left(\frac{2}{n}\right)} + \sqrt{f\left(\frac{3}{n}\right)} + \dots + \sqrt{f\left(\frac{n}{n}\right)} \right] =$$

- (A) $\frac{1 - \cos 1}{2}$ (B) $1 - \cos 2$ (C) $\frac{\sin 2}{2}$ (D) $\frac{1 - \cos 2}{2}$

Ans. D

Sol. -

Q23. Let $f(x)$ be a function satisfying $f(x) = f'(x)$ with $f(0) = 1$ and $g(x)$ be a function that satisfies

$$f(x) + g(x) = x^2. \text{ Then the value of the integral } \int_0^1 f(x)g(x)dx, \text{ is}$$

- (A) $e + \frac{e^2}{2} + \frac{5}{2}$ (B) $e - \frac{e^2}{2} - \frac{5}{2}$ (C) $e + \frac{e^2}{2} - \frac{5}{2}$ (D) $e - \frac{e^2}{2} - \frac{3}{2}$

Ans. D

Sol. -

Q24. The integral $\int \sqrt{\cot x} e^{\sin x} \sqrt{\cos x} dx$ equals

- (A) $\frac{\sqrt{\tan x} e^{\sqrt{\sin x}}}{\sqrt{\cos x}} + C$ (B) $2e^{\sqrt{\sin x}} + C$ (C) $-\frac{1}{2}e^{\sqrt{\sin x}} + C$ (D) $\frac{\sqrt{\cos x} e e^{\sqrt{\sin x}}}{2\sqrt{\cos x}} + C$

Ans. B

Sol. -

Q25. $\int_0^{\infty} \frac{x}{(1+x)(1+x^2)} dx =$

- (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) is same as $\int_0^{\infty} \frac{dx}{(1+x)(1+x^2)}$ (D) cannot be evaluated

Ans. A, C

Sol. -

Q26. Let $f(x) = \int_0^x e^t \cdot \sin(x-t) dt$ and $g(x) = f(x) + f''(x)$. Which of the following statements are correct?

- (A) $g(x)$ is positive $\forall x \in \mathbb{R}$. (B) $g(x)$ is a constant function (C) $g(1) = e$
(D) $g'(x) = g(x)$

Ans. A, C, D

Sol. -

Q27. Let $I_n = \int_0^{\sqrt{3}} \frac{dx}{1+x^n}$ ($n = 1, 2, 3, \dots$) and $\lim_{n \rightarrow \infty} I_n = I_0$ (say), then which of the following statement(s) is/are correct? (Given : $e = 2.71828$)

- (A) $I_1 > I_0$ (B) $I_2 < I_0$ (C) $I_0 + I_1 + I_2 > 3$ (D) $I_0 + I_1 > 2$

Ans. D, C, A

Sol. -

Q28. Let $I_n = \int_0^{\pi/4} (\tan x)^n$ and let $J_n = (-1)^n I_{2n}$ for $n = 0, 1, 2$ then which of the following hold(s) good?

- (A) $I_n + I_{n+2} = \frac{1}{n+1}$ (B) $J_n - J_{n-2} = \frac{(-1)^n}{2n-1}$ for $n \geq 1$
(C) If $u = \tan x$ then $I_n = \int_0^1 \frac{u^n}{1+u^2}$ (D) $\lim_{n \rightarrow \infty} J_n \rightarrow 0$

Ans. D, C, B, A

Sol. -

Q29. Which of the following statement(s) is/are true?

- (A) If function $y = f(x)$ is continuous at $x = c$ such that $f(c) \neq 0$ then $f(x) f(c) > 0 \forall x \in (c-h, c+h)$ where h is sufficiently small positive quantity.
(B) $\lim_{n \rightarrow \infty} \frac{1}{n} \ln \left(\left(1 + \frac{1}{n}\right) \left(1 + \frac{2}{n}\right) \dots \left(1 + \frac{n}{n}\right) \right) = 1 + 2 \ln 2$.

Let f be a continuous and non-negative function defined on $[a, b]$. If

(C) $\int_a^b f(x) dx = 0$ then $f(x) = 0 \forall x \in [a, b]$.

(D) Let f be a continuous function defined on $[a, b]$ such that $\int_a^b f(x) dx = 0$, then there exists atleast one $c \in (a, b)$ for which $f(c) = 0$.

Ans. D, C, A

Sol. -

Q30. Let $f(x) = \int_0^x e^{t-[t]} dt$ ($x > 0$), where $[x]$ denotes greatest integer less than or equal to x , is

(A) continuous and differentiable $\forall x \in (0, 3]$

(B) continuous but not differentiable $\forall x \in (0, 3]$ (C) $f(1) = e$ (D) $f(2) = 2(e - 1)$

Ans. D, B

Sol. -

Q31. If $a, b, c \in \mathbb{R}^+$ then $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{(k+an)(k+bn)}$ is equal to

(A) $\frac{1}{a-b} \ln \frac{b(b+1)}{a(a+1)}$ if $a \neq b$ (B) $\frac{1}{a-b} \ln \frac{a(b+1)}{b(a+1)}$ if $a \neq b$ (C) non existent if $a = b$

(D) $\frac{1}{a(1+a)}$ if $a = b$

Ans. D, B

Sol. -

Q32. If $\int_{-\alpha}^{\alpha} \left(e^x + \cos x \ln \left(x + \sqrt{1+x^2} \right) \right) dx > \frac{3}{2}$, then possible value of a can be :

(A) 1 (B) 2 (C) 3 (D) 4

Ans. D, C, B, A

Sol. -